

Aligning Small LLMs with GRPO for Strict JSON Generation

Rule-Based Reward Optimization for Schema-Conformant Output

Giuseppe Bellamacina

2026

Problem & Objective

- **Small LLMs** (<2B) frequently fail to produce valid, schema-conformant JSON
- **Common errors:** unclosed brackets, type violations, conversational filler mixed with output
- **Goal:** use GRPO with rule-based rewards to reach >90% Pass@1
- **No neural reward model:** dense, **deterministic**, interpretable signal

✖ Typical failure

```
Sure! Here's the JSON you asked for:  
{  "name": "Alice",  "age": 30,  
  "skills": ["Python", "ML" }  
// Missing ] and extra text
```

✔ Expected output

```
“`json {"name": "Alice", "age": 30,  
  "skills": ["Python", "ML"]}”`
```



7 Rule-Based Rewards

Additive composition with graduated partial credit. Avoids zero-advantage collapse inherent to hard gating



3-Stage Curriculum

Progressive difficulty scaling (35%→10% simple, 10%→65% hard) with decreasing temperature across 2500 steps



5 Models × 4 Strategies

Systematic comparison: 135M–2B parameters, Think/No-Think × Standard/Curriculum. Identical hyperparameters



Reproducible Infrastructure

Per-model YAML configs, SLURM chain pipeline, stratified eval, parametric dataset generator (24 templates)

Models & Setup

Model	Params	Arch
SmolLM2-135M-Instruct	135M	LLaMA-like
SmolLM2-360M-Instruct	360M	LLaMA-like
Qwen2.5-0.5B-Instruct	0.5B	Qwen2.5
TinyLlama-1.1B-Chat	1.1B	LLaMA 2
Gemma-2-2B-it	2B	Gemma 2

Configuration

 4-bit NF4 quantization

 LoRA $r=16$, $\alpha=32$

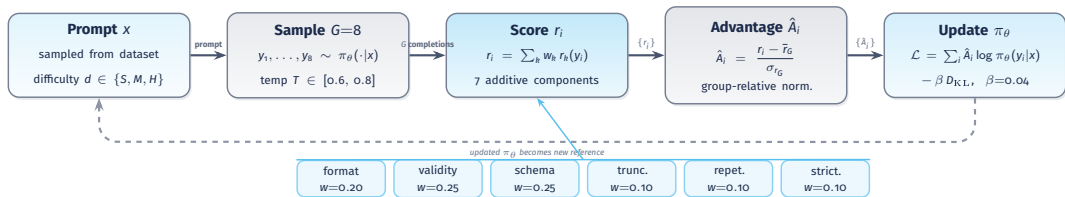
 1× NVIDIA L40S (48 GB)

 2500 steps / model

 Paged AdamW 8-bit

 LR 5×10^{-6} , cosine

Group Relative Policy Optimization (GRPO)



Rule-Based Reward System

Component	Range	Think	NoThink	What it measures
 format	[0, 1]	0.10	0.20	“‘ json “‘ fence → 1.0 generic fence → 0.5 none → 0.0
 validity	[0, 1]	0.15	0.25	JSON parseability; partial credit by error position (0.20–0.70)
 schema	[0, 1]	0.20	0.25	Keys, types, nesting depth, required fields
 reasoning	[−0.5, 1]	0.25	0.0	<think> block quality; penalises missing/copied
 truncation	[−1, 1]	0.10	0.10	Unclosed braces / unterminated strings
 repetition	[−1, 1]	0.10	0.10	Token loops, repeated lines, duplicate blocks
 strictness	[−1, 1]	0.10	0.10	Extra text outside JSON >50% of output

Design principle: Additive composition, no hard gating. Gradient is always non-zero; avoids [zero-advantage collapse](#). When Think is off, reasoning weight is redistributed proportionally.

Training Strategies & Dataset

S:35 M:55 H:10

Stage 1: Format

800 steps

S:15 M:35 H:50

Stage 2: Progressive

800 steps

S:10 M:25 H:65

Stage 3: Full

900 steps

	Standard	Curriculum
No-Think	✓	✓
Think	✓	✓

Dataset

 24 parametric templates
(8 per difficulty tier)

 1500 samples/stage $\times$ 3
= **4500** training

 Eval: **300** balanced
(100 S / 100 M / 100 H)

 No ground-truth JSON
reward evaluates directly



Experimental Results

5 models $\times$ 4 strategies $\times$ 2 500 steps

Baseline Pass@1 — Off-the-Shelf Models

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	39.00%	58.00%	39.00%	20.00%
SmolLM2-360M	77.67%	85.00%	86.00%	62.00%
Qwen2.5-0.5B	92.67%	95.00%	95.00%	88.00%
TinyLlama-1.1B	79.33%	89.00%	76.00%	73.00%
Gemma-2-2B	98.33%	100%	100%	95.00%

39% floor
SmolLM2-135M overall

98% ceiling
Gemma-2-2B overall

300-sample balanced eval (100 S / 100 M / 100 H). Config: No-Think / Standard.

Peak Performance — After 2 500 GRPO Steps

Model	Pre	Peak	Best Configuration	Δ
SmolLM2-135M	31%	90.00%	Think / Standard	+58.67 pp
SmolLM2-360M	78%	96.67%	No-Think / Standard	+19.00 pp
Qwen2.5-0.5B	92%	98.00%	Think / Curriculum	+6.33 pp
TinyLlama-1.1B	79%	99.33%	No-Think / Curriculum	+20.00 pp
Gemma-2-2B	96%	97.67%	Think / Curriculum	+1.33 pp

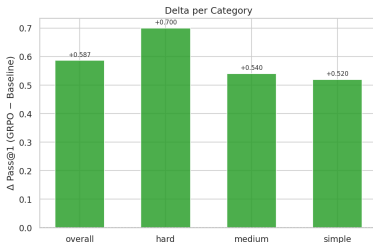
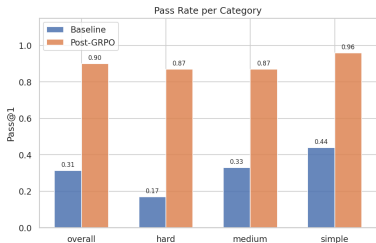
Pre = same-config baseline (Think prompting lowers small-model scores by ~ 8 pp vs. No-Think)

87 – 99% convergence band

All 5 models reach this band regardless of initial capability

Greatest Transformation — SmolLM2-135M (+58.67 pp)

Baseline vs Post-GRPO — SmolLM2-135M-Instruct



31%

Baseline

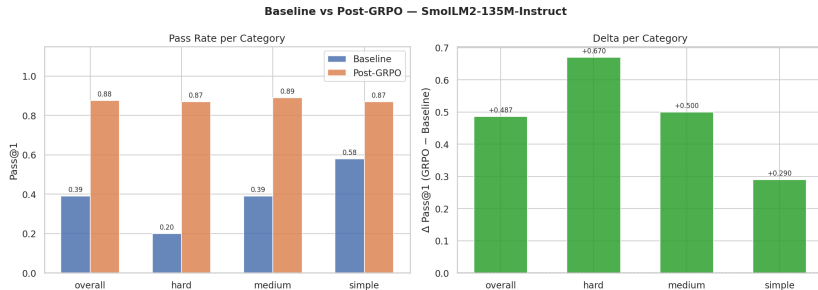


90%

Post-GRPO

Think / Standard
Small models learn to
exploit reward signals
without generating <think>

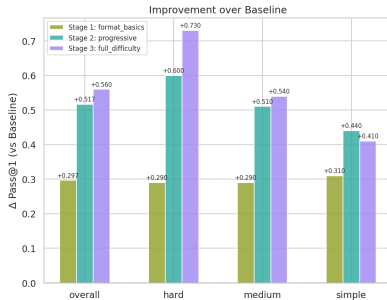
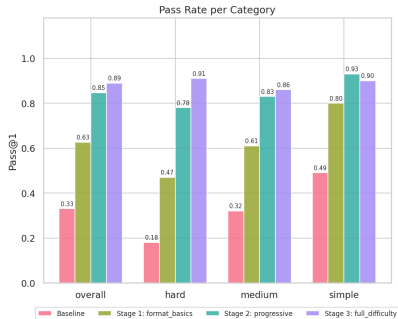
SmolLM2-135M — No-Think / Standard Comparison



Pass@1 by difficulty tier, before and after 2 500 GRPO steps (No-Think / Standard)

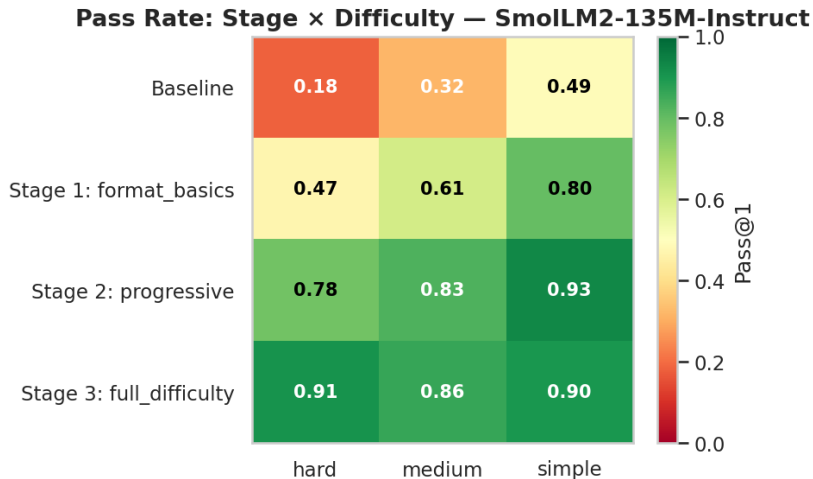
SmolLM2-135M — Curriculum Progression (No-Think)

Curriculum Progression — SmolLM2-135M-Instruct



33% → 62.7% → 84.7% → 89.0% across 3 curriculum stages

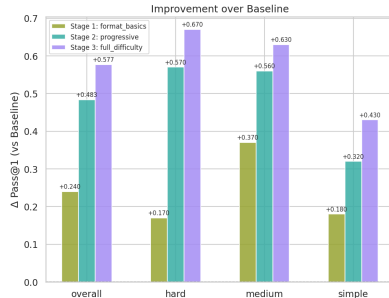
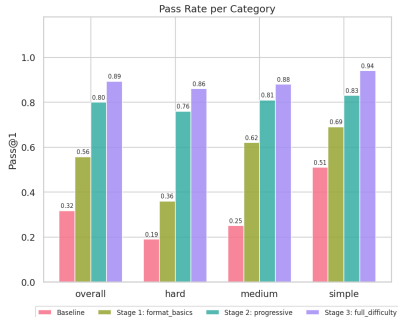
SmolLM2-135M — Difficulty Heatmap (No-Think / Curriculum)



Pass@1 per difficulty tier at each curriculum stage

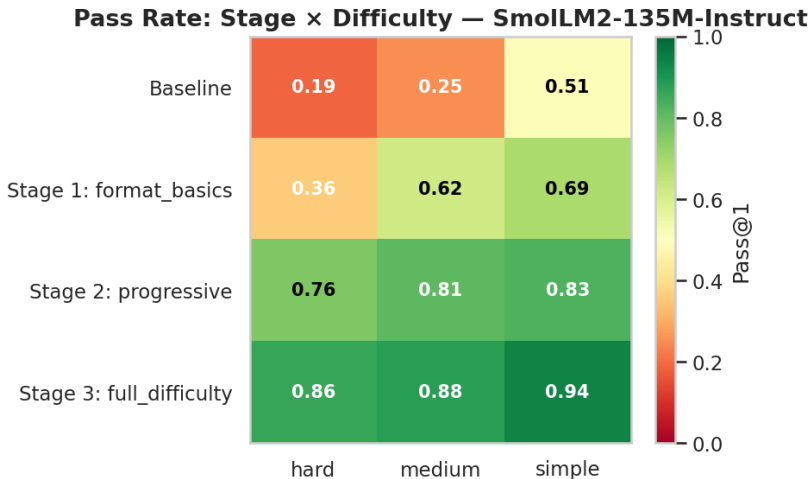
SmolLM2-135M — Curriculum Progression (Think)

Curriculum Progression — SmolLM2-135M-Instruct



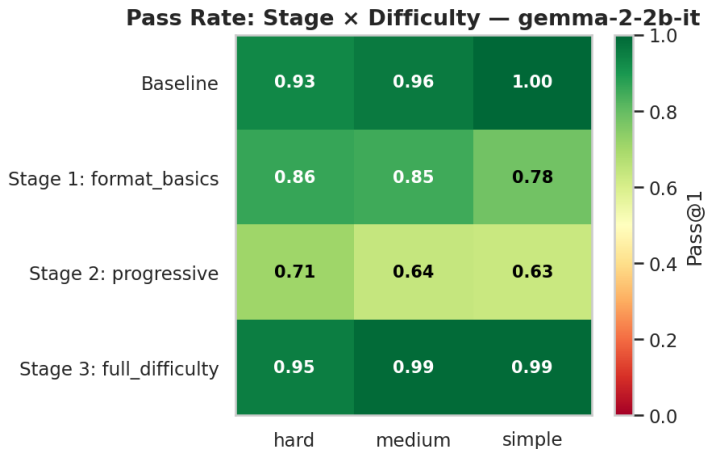
31.7% → 55.7% → 80.0% → 89.3%. Small models never generate <think> tokens but still exploit reward signals

SmolLM2-135M — Difficulty Heatmap (Think / Curriculum)



Per-difficulty breakdown reveals hard tier drives largest improvements

Anomaly — Gemma-2-2B U-Shaped Learning Curve



Temporary collapse

96% → 83% → 66% → 98%

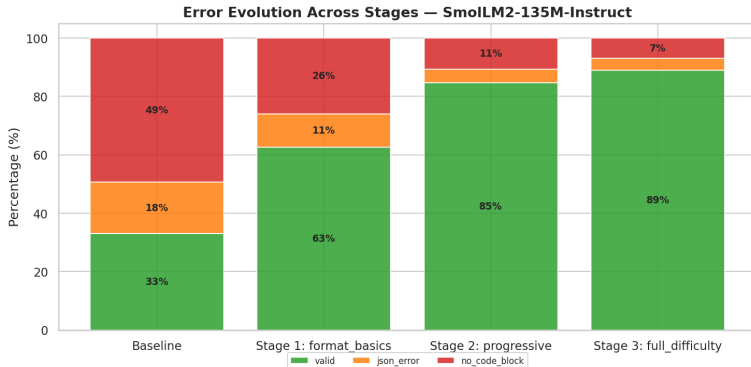
Forcing <think> on trivially simple schemas temporarily confuses alignment.

Full recovery once hard schemas dominate in Stage 3.

Key insight

Curriculum pacing must match model capacity. Strong models need less “warm-up”.

Error Evolution — How Failures Disappear



Stage 1

Dominant error:
`no_code_block`
Model learns the
“`json`” fence first

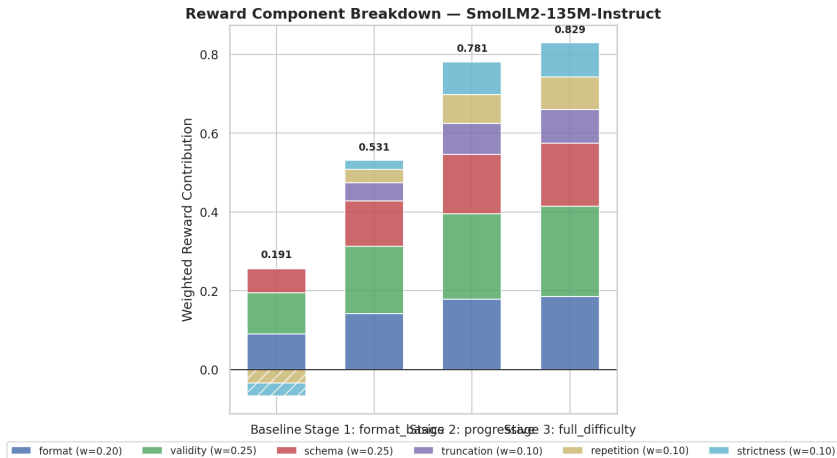
Stage 2

Format mastered, residual:
`json_error`
Focus shifts to structure

Stage 3

Near-zero errors remaining
Only edge-case schema fail-
ures

Reward Breakdown — SmolLM2-135M



Component contributions at final checkpoint (No-Think / Curriculum)



Think vs. No-Think

Small models (135M, 360M, 0.5B) never generate `<think>` tokens; only Gemma and TinyLlama can. 135M peaks **90%** in Think config but by exploiting other rewards, not reasoning. Think doesn't help for these tasks.



Curriculum = Safety Net

Standard causes Gemma-2-2B to regress (-1.67pp). Curriculum flips this to $+0.67\text{pp}$. Produces the absolute highest scores (TinyLlama 99.33%, Qwen 98.00%).



Inverted-U Gain Pattern

Smallest models gain most: 135M $+58\text{pp}$. Mid-range (360M, 1.1B): $+13\text{--}26\text{pp}$. Already-strong (0.5B, 2B): $+0\text{--}8\text{pp}$, approaching ceiling.



Hard Tier Drives Improvement

Baselines: 17–20% hard pass for 135M. Post-GRPO: **86–99%** on hard prompts. Simple saturates to 100% very early (Stage 1).

✓ Technical Lessons

- **Additive rewards** > **gating**

Zero-advantage collapse if all tie → vanishing gradient

- **Graduated partial credit**

Binary validity gives no signal for near-valid JSON

- **4-bit NF4 + LoRA** sufficient

<2% params trained, single GPU, identical hyperparams

- **Weight redistribution**

Auto-preserved component ratios when Think off

! Limitations

- JSON only — not directly generalizable to other formats
- Fully synthetic dataset; potential real-world gap
- No comparison with DPO/PPO (only GRPO tested)
- Fixed hyperparameters — per-model tuning could help

Conclusions

✓ GRPO + rule-based rewards brings all 5 models to 87–99% Pass@1

👤 **Standard** is optimal for weak models:

SmolLM2-135M **+58.67 pp**

🔒 **Curriculum** is a safety net for strong models:

Gemma-2-2B: -1.67 (std) $\rightarrow +0.67$ (curr)

🧠 **Think mode**: <2 pp marginal gain;

small models can't generate `<think>`, not useful for this task

⚙️ **Reward design** $>$ **hyperparameter tuning**:

identical config works for 135M – 2B